Cost-Aware LLM Routing with Online Reinforcement Learning

Ravi Bhavana Prof. Manisha Padala

Indian Institute of Technology Gandhinagar

INTRODUCTION

Overview: For each query, which of five open language models should answer it? I build a router that learns *online* and compare it with four published routers.

WHY THIS MATTERS

- Large models answer well but cost more; small models are cheaper but weaker
- Published routers train *offline* on the models' answers to every training query
- An online router needs feedback only on the answer it served

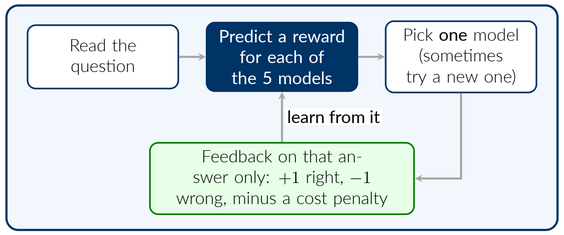
THE SETUP

Model	Params	Cost
flan-t5-small	80M	0.011
flan-t5-base	250M	0.036
flan-t5-large	780M	0.111
flan-t5-xl	3B	0.429
Llama-2-7b-chat	7B	1.000

Cost = parameters / 7B; test cost = average over queries of the model picked.
Data (train / validation / test): MMLU 12,542 / 1,500 / 1,531 questions (accuracy); MixInstruct 9,000 / 1,000 / 1,500 queries (pass rate = BARTScore ≥ -4.55).
Utility = (+1 right, -1 wrong) $-$ 0.5 $\times$ cost. Higher is better.
λ (cost weight) is chosen on the validation split; results are on the test set.

MY ONLINE RL ROUTER

ONLINE Learns by trial and error from feedback on **one** answer per question.



The question becomes a DeBERTa embedding reduced to 65 numbers. A small network predicts one reward per model; only the picked model's prediction is updated.

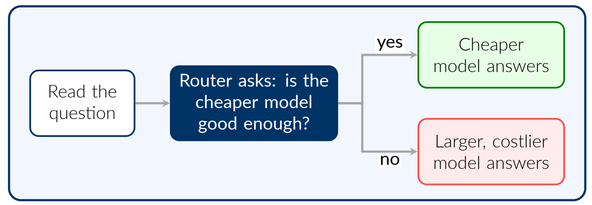
EXPLORATION RULES TRIED

- ε-greedy: usually the best model, sometimes a random one
- **Boltzmann:** pick in proportion to predicted reward
- **LinUCB:** a bonus for models it knows little about
- **Thompson sampling:** act on a sampled guess of each model's quality

Outcome: seven variants (20 seeds) landed within 0.03 utility of each other, so no rule is clearly better. The tables use the rule chosen on validation: LinUCB (MMLU), Thompson sampling (MixInstruct); fine-tuned runs use Boltzmann.

HYBRIDLLM (ICLR 2024)

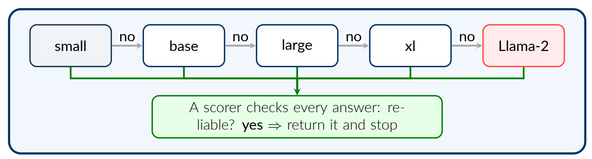
OFFLINE Trained once on the quality gap between a cheaper and a costlier model.



One pair at a time; a threshold sets the share sent to the cheaper model. I ran three pairs (costlier model = Llama-2): **large gap** (small + Llama-2), **medium gap** (large + Llama-2), **small gap** (xl + Llama-2).

FRUGALGPT (TMLR 2024)

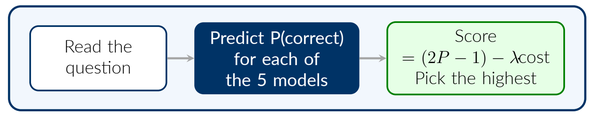
OFFLINE A cascade: a scorer decides whether to accept or escalate.



The cheapest model answers first; if the scorer doubts it, the question moves up a model. Cost counts every model that ran. Compared on MMLU; also reproduced on the paper's HEADLINES dataset.

PREDICTIVE ROUTER

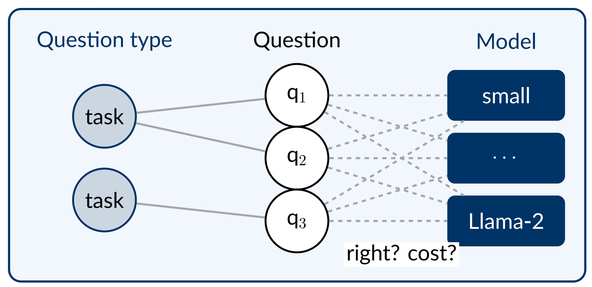
OFFLINE Trained beforehand on the right-or-wrong answers of **all five** models.



A small neural network (65-128-64-5); λ is used only when routing. Inspired by the predictive routers of RouterBench (Hu et al., 2024).

GRAPHROUTER (ICLR 2025)

OFFLINE Trained beforehand on all five models' answers.



Training questions link to the models' results (right or wrong, cost). A new question links to its type; nodes pass messages along the lines (2 rounds); each model gets a score and the highest wins. Mine is a simplified re-implementation.

RESULTS: MMLU (TEST SET)

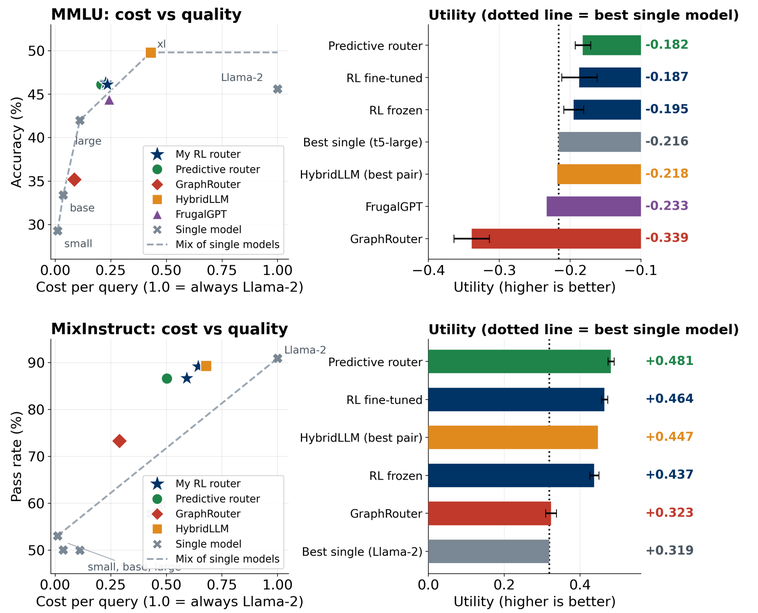
Method	Cost	Accuracy	Utility
Best single (flan-t5-large)	0.111	42.0%	−0.216
RL, frozen, LinUCB (20 seeds)	**0.234**	**46.1%**	−0.195
RL, fine-tuned, Boltzmann (3 seeds)	**0.226**	**46.3%**	−0.187
HybridLLM (best pair: xl + Llama-2)	0.429	49.8%	−0.218
FrugalGPT (confidence scorer)	0.242	44.4%	−0.233
Predictive router (5 seeds)	0.207	46.1%	−0.182
GraphRouter (5 seeds)	0.085	35.2%	−0.339

RESULTS: MIXINSTRUCT (TEST SET)

Method	Cost	Pass rate	Utility
Best single (Llama-2)	1.000	90.9%	+0.319
RL, frozen, Thompson (20 seeds)	**0.592**	**86.7%**	+0.437
RL, fine-tuned, Boltzmann (3 seeds)	**0.643**	**89.2%**	+0.464
HybridLLM (best pair: small + Llama-2)	0.678	89.3%	+0.447
Predictive router (5 seeds)	0.502	86.6%	+0.481
GraphRouter (5 seeds)	0.288	73.3%	+0.323

Utility = (2 × quality − 1) − 0.5 × cost. Best pair = best utility among the three HybridLLM pairs. HybridLLM and FrugalGPT are single runs; FrugalGPT was not run on MixInstruct.

HOW THE METHODS COMPARE



Left: cost vs quality (up and left is better). Right: utility; the dotted line is the best single model. Error bars: spread over seeds.

KEY FINDINGS

1. **RL beats the best single model.** Utility gain: **+0.118** on MixInstruct, and a smaller +0.020 on MMLU (frozen bandit, 20 seeds).
2. **It cuts cost and keeps most of the quality.** Versus the most accurate model (fine-tuned RL): 47% cheaper for -3.5 points on MMLU, 36% cheaper for -1.7 points on MixInstruct.
3. **At similar cost it matches the published routers.** MMLU: 46.3% at 0.226 vs FrugalGPT 44.4% at 0.242. MixInstruct: same quality as HybridLLM (89.2% vs 89.3%) at 5% lower cost.
4. **Offline routers can score a little higher, but learn at a higher cost.** The predictive router ties RL on MMLU and leads by 0.017 to 0.044 utility on MixInstruct; RL needs 2.5 to 6 times less compute and feedback on one answer only.

COST OF LEARNING

Method	Parameter-calls / query
My RL router	1.8B to 4.4B
HybridLLM	7.1B to 10.0B
Predictive, Graph, Frugal	11.1B

LIMITATIONS & FUTURE WORK

- Feedback is simulated from precomputed answers; test **real user feedback**.
- Try more advanced online learning algorithms in place of ε-greedy, Boltzmann, LinUCB and Thompson sampling.
- Only five open models were tested; try **paid models such as GPT**, where cost is a per-token price instead of parameters.
- HybridLLM and FrugalGPT are single runs; repeat them with several seeds.

REFERENCES

1. Ding et al. (2024). Hybrid LLM. *ICLR*.
2. Chen, Zaharia & Zou (2024). FrugalGPT. *TMLR*.
3. Hu et al. (2024). RouterBench. *arXiv:2403.12031*.
4. Feng, Shen & You (2025). GraphRouter. *ICLR*.
5. Li et al. (2010). LinUCB. *WWW*.
6. Agrawal & Goyal (2013). Thompson sampling. *ICML*.

ACKNOWLEDGEMENT

I sincerely thank my supervisor Prof. Manisha Padala for guidance and support.